DO NOW — RECALL

Lesson 056 · Tree Traversals: Post-order and Breadth-fi

Name: _____ Class: _____ Date: _____

1. What is a leaf node?

2. What is BFS's underlying structure?

3. What order does postorder use?

4. A binary tree has root 10, children 5 and 15, and node 5 has right child 7 only. Trace its postorder traversal, using left subtree, right subtree, root.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Tree node with no children.

2. Model answer

A queue.

3. Model answer

Left subtree, right subtree, root.

4. Model answer

7, 5, 15, 10. Finish the left subtree, then the right subtree, then visit the root.

By the end, you can…

• Post-order visits both child subtrees before its parent. Traverse left subtree, right subtree, then root.

• Breadth-first visits by levels, left-to-right with a queue that stores the next vertices.

• Traversal result depends on the precise tree and stated child order. BFS queue changes must be shown to justify an answer.

Vocabulary — know these terms

postorder — Left-right-root tree traversal.

BFS — Breadth-first level exploration.

queue — FIFO worklist.

recursive — Process invoking itself on children.

NEW LEARNING

Read, annotate and explain

Understand

Post-order visits both child subtrees before its parent. Traverse left subtree, right subtree, then root.

Model

Breadth-first visits by levels, left-to-right with a queue that stores the next vertices.

Evaluate

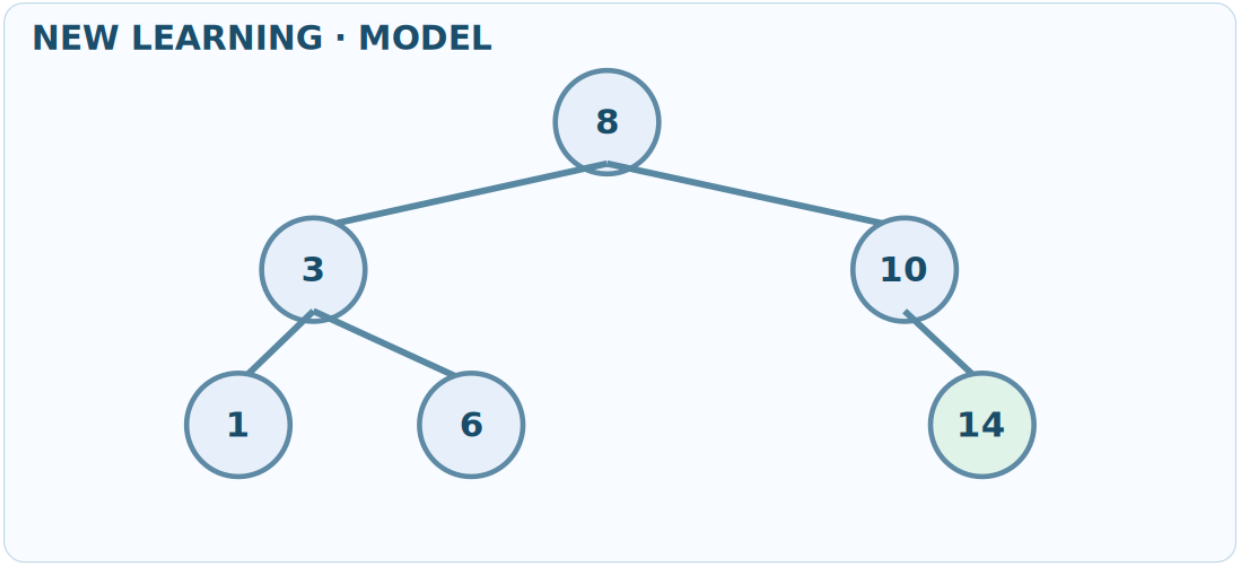
Traversal result depends on the precise tree and stated child order. BFS queue changes must be shown to justify an answer.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



NEW LEARNING · MODEL: 8; 3; 10; 1; 6; 14

I DO — WORKED EXAMPLE

Follow the worked method

Give postorder for tree root5, left2, right7; node2 has left1.

Explain

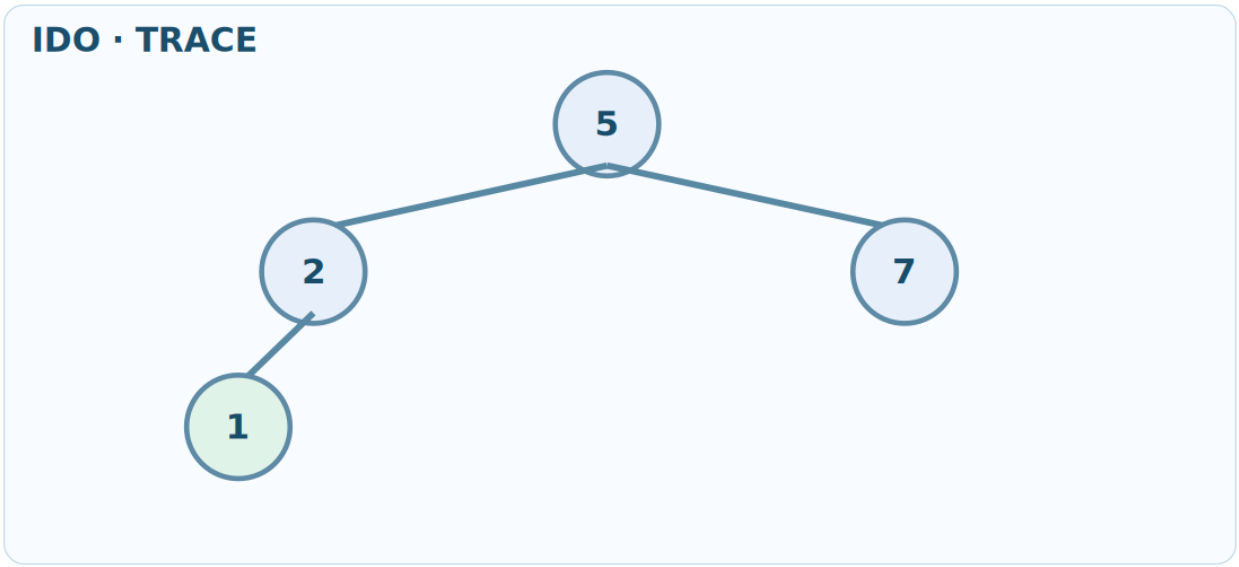
PostOrder(5) waits for PostOrder(2), which prints 1 then 2. The right call prints 7; only then is 5 printed: 1,2,7,5. null returns without printing. Node supplies Key, Left and Right.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Give postorder for tree root5, left2, right7; node2 has left1.

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IDO · TRACE: 5; 2; 7; 1

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Give breadth-first order for root9, left4, right12; 4 has right6.
Source code / input:
Queue starts [9] → enqueue [4,12] → visit4/enqueue6 → visit12 → visit6.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Root8, children3 and10, with 3 children1 and6, 10 right child14. Give postorder. [6]

Question 2

Give breadth-first order for same tree, and first two queue changes. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Explain why postorder is useful for deleting tree structures but BFS for shortest unweighted-edge distance. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

Breadth-first tree traversal uses a stack to process deepest nodes first.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

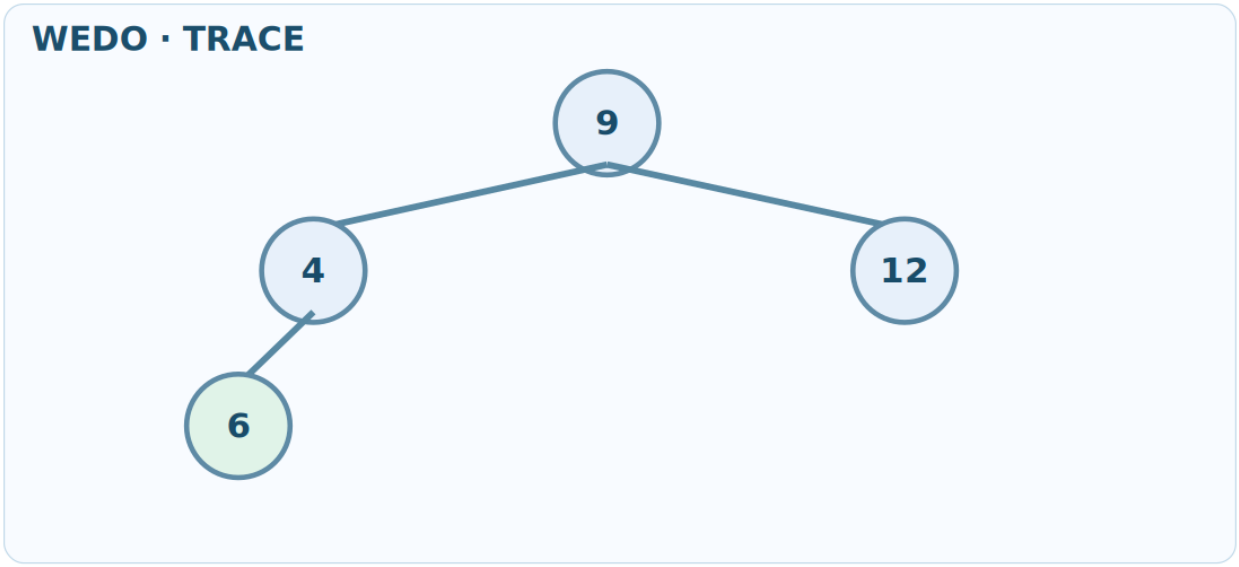
Explain
Order 9,4,12,6. Visit by depth, using FIFO.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Give breadth-first order for root9, left4, right12; 4 has right6.

Order 9,4,12,6. Visit by depth, using FIFO.



WEDO · TRACE: 9; 4; 12; 6